Name :- B. Satya Hyndavi

Course Name:- Artifical Intilligence.

Reg NO :- 192311178.

Cours code:- CSA1704

182
20

(1) Smart Home Automation System using intelligent Agent

Introduction:- A smart automation system controls lighting, temperature, and security by using sensor inputs such as motion detection, temperature, and time of day. Intelligent agents helps the system make decision automatically

Types of Intelligence Agents:-

(1) simple Reflex Agent:-

* Acts only on the current input
* Uses conditional_action rules
* Does not remember Perious events

Ex:- if motion is detected - Turn ON the lights

Advantages:-

* Fast
* Easy to implement

Disadvantages

* Cannot learn user preferences
* Cannot handle complex situation

2) Model - Based Agent

* Maintain an internal model of the environment
* Uses both current & past information

ex:- Camera, Thermostat

Advantages

* Better decision making
* Handles partially observable enviornments

Disadvantages

* More mer̶ory & computation required

(3) Goal-Based Agent:-

* Makes decision to achieve specific goals

Ex:- Temperature, Security

Advantages

* Flexible

Disadvantages

* Requires planning
* Slower than reflex agent

(4) Utility-Based Agent

* Selects the action that provides maximum benefit
* Considers comfort, energy saving, & security together

Ex:- comfort, Energy

Advantages

* Optimizes overall performance
* Balance multiple objectives.

Disadvantages:-

* Designing the utility function is difficult

A hybrid Agent (Model - Based + Utility) is most effective.

* Learns user perference.
* optimize comfort
* Saves energy
* Imporves security.
* Makes intelligent decision in real time

(2) Robat Navigation in an Unknown Maze

Introduction!-

A robot is an unknown maze uses Search Algorithm to find the exit without knowing the path in advance.

Uniform cost Search

Expands the path with the lowest cost first

ex:-

GPS, Maps

Advantages

It finds lowest path.

Dis advantages

uses more memory

Iterative Deepening Search!-

performs Depth - first search repeatedly with increasing depth limits

ex!- puzzle, Maze.

Advantages :-

* Less memory.

Disadvantages :-

* Repeats searching.

Conclusion

The Goal-Based + UCS.

It helps to find lowest path to the exit.

Here the

* Time complexity of Uniform Cost search

Time :- $O(b^{(1+C^*/\varepsilon)})$

Space :- $O(b^{(1+C^*/\varepsilon)})$

* Time complexity of Iterative Depending Search

Time :- $O(b^d)$

Space :- $O(bd)$

Diagram :-